UNIT 4 · 9 hours

BJT as a Switch



Transistors

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

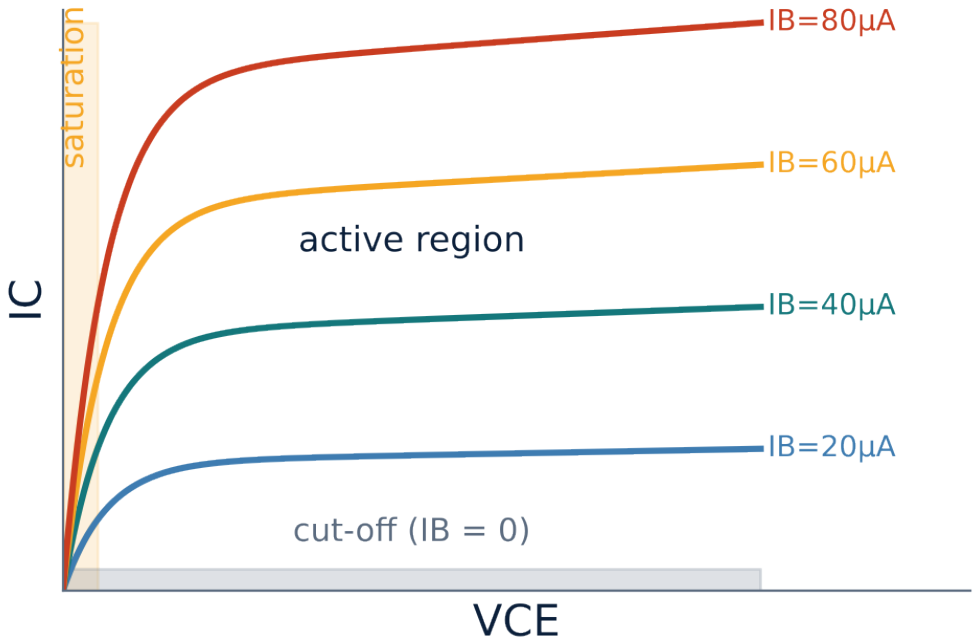
- Identify cut-off, active and saturation regions
- Explain how a BJT acts as an open/closed switch
- Design base resistance for saturation
- Apply the switch to drive an LED/relay

Syllabus context

Construction and operation, BJT as a switch, working and characteristics of n-channel enhancement type MOSFET, common source amplifier, DC load line analysis, MOS inverter.

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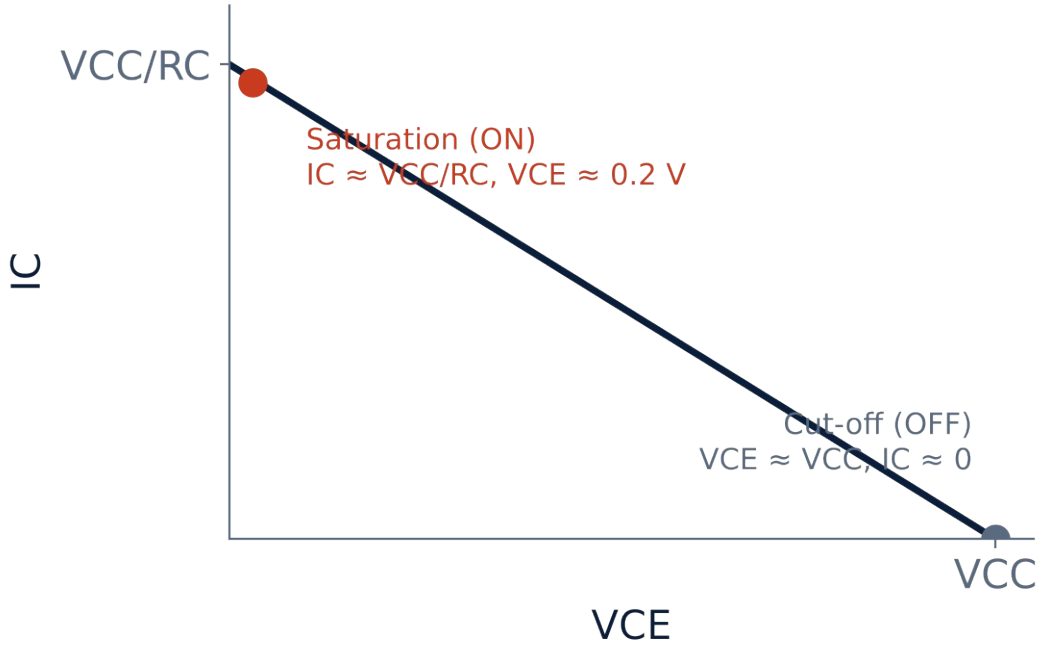
Output characteristics and regions



- Cut-off: IB = 0, IC ≈ 0 (junctions reverse biased).
- Active: IC = β IB; used for amplification.
- Saturation: both junctions forward biased; VCE(sat) ≈ 0.2 V.
- Switch uses only cut-off and saturation.

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DC load line and the two switch states



- Load line: VCE = VCC − IC RC.
- OFF: VCE ≈ VCC, IC ≈ 0 (open switch).
- ON: IC ≈ VCC/RC, VCE ≈ 0.2 V (closed switch).

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OFF state versus ON state

OFF: cut-off

- VBB (input) = 0 V, so IB = 0
- Both junctions reverse biased
- IC ≈ 0: open switch
- VCE ≈ VCC; LED off

ON: saturation

- Input high; IB large enough
- Both junctions forward biased
- IC(sat) ≈ VCC/RC: closed switch
- VCE(sat) ≈ 0.2 V; LED on

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Design for saturation

$$I_{C(sat)} \approx (V_{CC} - V_{CE(sat)}) / R_C$$

load current

$$I_{B(min)} = I_{C(sat)} / \beta$$

minimum base current

$$R_B = (V_{in} - V_{BE}) / I_B$$

choose $I_B \geq 2–5 \times I_{B(min)}$

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Numerical: transistor switch design

Problem VCC = 10 V, RC = 1 kΩ, β = 100, Vin = 5 V (high). Find IC(sat), IB(min) and RB for IB = 2 × IB(min).

- IC(sat) = (10 − 0.2)/1 kΩ = 9.8 mA
- IB(min) = 9.8 mA/100 = 98 µA; IB = 196 µA
- RB = (5 − 0.7)/196 µA

Answer: RB ≈ 22 kΩ

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LED switch application

- A 2 s square wave on the input: LED on for 1 s, off for 1 s.
- Input low (0 V): transistor in cut-off, no collector current, LED off.
- Input high: transistor saturates, collector current flows through the LED, LED on.
- Other uses: relay/motor drivers, digital logic gates, inverters.

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Key takeaways

- Switch uses cut-off (OFF) and saturation (ON)
- ON: IC ≈ VCC/RC, VCE(sat) ≈ 0.2 V; OFF: IC ≈ 0, VCE ≈ VCC
- Design IB ≥ IC(sat)/β with overdrive
- Used in LED/relay drivers and digital logic

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Practice questions

Q1. Explain the operation of a BJT as a switch with a diagram.

Q2. Find RB for VCC = 5 V, RC = 500 Ω, β = 50, Vin = 5 V, overdrive 2.

Q3. State the conditions for cut-off and saturation.

Q4. Why must the transistor not be left in the active region when used as a switch?